

In federated-learning systems, how do empirical studies quantify the effectiveness of differential privacy against membership inference or canary-audited membership leakage, and how do results vary by privacy unit (record versus client), adversary access to model updates, and real-world application domain? Please report the abstract of the most relevant papers.

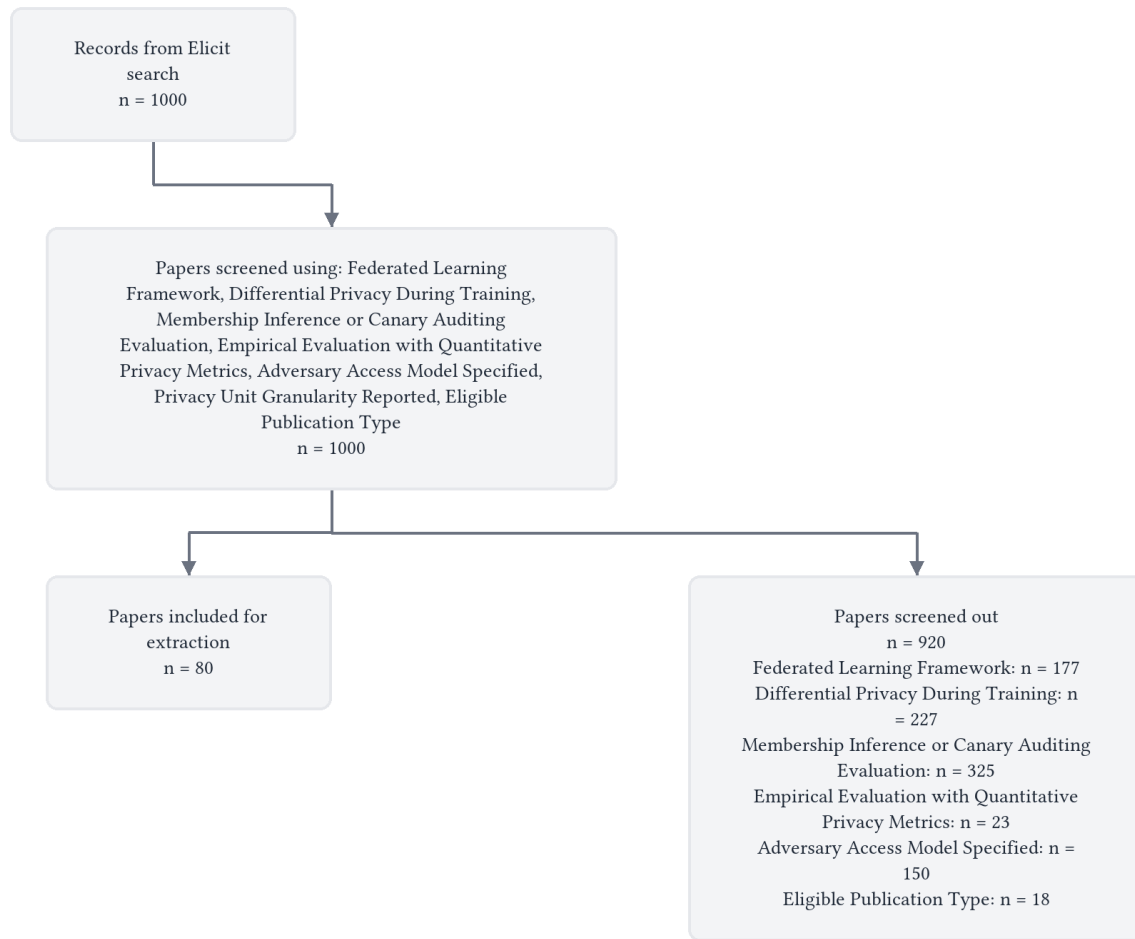
Empirical studies consistently show that differential privacy reduces membership inference attack success to near-chance levels against passive adversaries at moderate privacy budgets ($\epsilon \leq 2-5$), but protection degrades substantially against active adversaries who manipulate model updates, tightens in cost when the privacy unit is elevated from record-level to client- or subject-level, and varies by domain — with healthcare applications achieving the most favorable privacy-utility trade-offs and high-dimensional or interaction-rich domains such as recommendation systems and large language models requiring either tighter budgets or complementary defenses to achieve comparable protection.

Abstract

Across 80 empirical studies spanning healthcare, computer vision, NLP, IoT, and specialized domains, differential privacy (DP) consistently reduces membership inference attack (MIA) success in federated learning, but the degree of protection varies substantially by adversary capability, privacy unit, and application context. Against passive or black-box adversaries evaluating final global models, moderate DP budgets ($\epsilon \leq 2-5$) routinely drive MIA performance to near-chance levels (AUC 0.50–0.53), as demonstrated in COVID-19 CT prognostication [1], clinical AI on MIMIC-IV [2], maritime vessel classification [3], ERP fraud detection [4], and synthetic EHR generation [5], typically with utility losses of 1–5 percentage points. Against active adversaries who manipulate model parameters during training, however, DP provides only partial mitigation: active server-side attacks maintained ≥ 0.77 success under LDP at $\epsilon \geq 5$ [6], and active attacks on large language models achieved 100% success without DP, declining only below AUC 0.72 at $\epsilon = 5$ [7]. Canary-based empirical auditing reveals that formal DP bounds overestimate actual leakage by 4–5 times [8], suggesting theoretical guarantees are conservative under realistic threat models.

Privacy-unit granularity is a critical moderator: within a single FEMNIST evaluation, item-level DP ($\epsilon = 4.0$) reduced attack F1 from 0.83 to 0.76 while preserving 85.1% task accuracy, whereas user-level DP achieved stronger MIA reduction (F1 = 0.67) but at the cost of accuracy collapsing to 41.0% [9]. Subject-level DP offered an intermediate trade-off (F1 = 0.72, accuracy 81.5%) [9], though subject-level attacks remained effective in some configurations even with DP [10]. Domain-specific patterns are evident: healthcare applications generally achieve favorable privacy-utility trade-offs due to lower model complexity [1, 2, 11], while federated recommendation systems show that LDP is ineffective against interaction-level MIA unless recommendation quality is largely sacrificed [12], and high-dimensional vision tasks require tighter budgets or complementary defenses. Combining DP with secure aggregation, homomorphic encryption, or architectural defenses such as model splitting or layer mixing yields the strongest empirical protection, reducing MIA success by over 84% in some configurations [13, 14]. Taken together, the evidence indicates that DP is a necessary but context-dependent safeguard whose effectiveness must be evaluated against the specific adversary model, privacy unit, and domain rather than assumed from formal guarantees alone.

Flow Diagram



Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "In federated-learning systems, how do empirical studies quantify the effectiveness of differential privacy against membership inference or canary-audited membership leakage, and how do results vary by privacy unit (record versus client), adversary access to model updates, and real-world application domain? Please report the abstract of the most relevant papers."

The search returned 1,000 total results from Elicit.

We retrieved 1000 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Federated Learning Framework:** Does the study take place within a genuine federated learning (FL) framework (e.g., FedAvg, FedProx, cross-silo, cross-device FL) involving at least two clients that perform local training and communicate model updates (gradients or parameters) to a server or aggregator — as opposed to

centralized ML, split learning without FL aggregation, purely peer-to-peer settings, federated databases/query systems, or single-node simulations that partition data without a real client-server communication protocol?

- **Differential Privacy During Training:** Does the study implement or evaluate at least one formal differential privacy (DP) mechanism (e.g., DP-SGD, Gaussian mechanism, Laplace mechanism, local DP, shuffling-based DP, amplification-by-sampling) applied to the FL training process itself (e.g., gradient or parameter perturbation, local randomization during training) — rather than relying solely on non-DP privacy tools (e.g., secure aggregation, homomorphic encryption, gradient clipping alone) or applying DP only to final model outputs or predictions after training?
- **Membership Inference or Canary Auditing Evaluation:** Does the study evaluate privacy leakage using membership inference attacks (MIA) — including passive or active adversaries — or canary/secret-sharer auditing techniques that empirically measure membership leakage, rather than exclusively evaluating other privacy threats such as gradient inversion, attribute inference, data poisoning, or model stealing without any MIA or canary-based component?
- **Empirical Evaluation with Quantitative Privacy Metrics:** Does the study report empirical, experimental results with at least one quantitative privacy metric (e.g., attack success rate, AUC of MIA, empirical epsilon, adversary advantage, true/false positive rates, privacy loss bounds) — as opposed to being a purely theoretical or proof-only paper, or a study that discusses DP and MIA only qualitatively without any numerical results?
- **Adversary Access Model Specified:** Does the study explicitly describe or clearly allow inference of the adversary's access model (e.g., white-box access to model updates or gradients, black-box access to model outputs only, honest-but-curious server, malicious clients, passive observer)?
- **Privacy Unit Granularity Reported:** Does the study specify or clearly allow inference of the privacy unit used (e.g., per-sample/record-level DP versus per-client/user-level DP)?
- **Eligible Publication Type:** Is the study published as a peer-reviewed conference paper, journal article, preprint (e.g., arXiv), systematic review, or meta-analysis — rather than being a narrative survey, opinion piece, or position paper that summarizes existing work without contributing new empirical experiments or systematic synthesis?

Papers that failed any criterion were automatically excluded; all other papers, including those with a Maybe on any criterion, were screened in.

At abstract screening, the number of papers excluded for each primary reason was:

- **Federated Learning Framework:** $n = 177$
- **Differential Privacy During Training:** $n = 227$
- **Membership Inference or Canary Auditing Evaluation:** $n = 325$
- **Empirical Evaluation with Quantitative Privacy Metrics:** $n = 23$
- **Adversary Access Model Specified:** $n = 150$
- **Eligible Publication Type:** $n = 18$

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Main findings:**

Summarize the results or conclusions of the study. Use bullet points (each starting with a dash). Each bullet point should consist of only one concise sentence. Give a minimum of one bullet point and a maximum of three. Make sure they convey the most important takeaways from the paper. Avoid being redundant.

- **Intervention:**

List all interventions that at least some participants received, but do not list controls or placebos. If only some participants received a certain intervention, note that. Note the frequency, duration, and amount or dose of the intervention. Be as precise as possible: if duration, frequency or dose are mentioned, they need to be in the answer. If there are multiple components to the intervention (e.g. a drug plus therapy), state all of them.

- **Outcome measured:**

There may be multiple primary outcomes or endpoints. If so, include all of them, as long as they are identified as main or primary outcomes or endpoints. Do not include secondary outcomes or endpoints. Include units if possible. For instance, if a study investigated the effects of caffeine on heart rate at rest, measured in bpm, and also measured as a secondary outcome the effect on anxiety, the answer is 'heart rate at rest (bpm)'. If the study investigated the effects of caffeine on heart rate, anxiety, and blood pressure, all of which seem to be equally important, then the answer is 'heart rate, anxiety, and blood pressure'. Be as precise as possible.

- **Study design:**

List all characteristics of study design, such as whether it was randomized, double-blind, controlled, placebo-controlled, non-controlled, multi-site, retrospective, stratified, crossover design, parallel design, an observational study, a meta-analysis, a systematic review, etc. Be comprehensive. If the study design is not mentioned, leave the answer blank.

Results

Characteristics of Included Studies

The review encompasses 80 sources published between 2018 and 2026, spanning computational experiments, benchmark evaluations, framework proposals, and deployed production systems. The majority are primary empirical studies that simulate or emulate federated-learning environments and measure membership-inference attack (MIA) success with and without differential privacy (DP). A smaller subset introduces novel attack or defense methodologies, proposes theoretical privacy frameworks, or evaluates DP in production deployments. Full-text retrieval was possible for 64 of the 80 sources; the remaining 16 were assessed from abstracts alone.

Study	Full text retrieved?	Study type	Application domain	DP variant(s)	Privacy unit	Adversary model
Taslima Khanam et al., 2026	No	Primary experiment [15]	EEG-BCI motor imagery [15]	DP on client updates, $\epsilon = 1, 5, 10$ [15]	Not specified [15]	Record-, feature-, gradient-, identity-level MIA [15]
Truc Nguyen et al., 2023	Yes	Primary experiment + theory [6]	Image classification (CIFAR-10, ImageNet, CelebA) [6]	LDP (OME, BitRand), $\epsilon = 1-10$ [6]	Record-level [6]	Active dishonest server [6]
A. M & C. L. Chowdhary, 2026	Yes	Controlled FL simulation [16]	Medical MRI classification [16]	Adaptive per-parameter DP, $\epsilon = 0.5-5.0$ [16]	Client-level [16]	Black-box MIA [16]
Samuel Maddock et al., 2022	Yes	Canary-audit framework [8]	Image (CIFAR-10, CelebA) and language (Sent140, Shakespeare) [8]	DP-SGD with clipping [8]	Record-level (canary) [8]	Strong canary adversary [8]
Adiah Qazi et al., 2026	Yes	Federated ensemble evaluation [4]	ERP fraud detection [4]	FS-CDP + DP-SGD, $(\alpha, \epsilon) = (8, 1.0)$ RDP [4]	Record-level [4]	Shadow-model MIA, LiRA [4]

Study	Full text retrieved?	Study type	Application domain	DP variant(s)	Privacy unit	Adversary model
Anshuman Suri et al., 2022	Yes	Multi-configuration attack study [9]	Handwriting (FEMNIST), text (Shakespeare), synthetic [9]	Item-, user-, subject-level DP, $\epsilon = 0.5-4.0$ [9]	Record, user, and subject [9]	Black-box distribution-based and item-based [9]
Hongkyu Lee et al., 2021	No	Primary experiment [17]	Not specified (FL benchmark) [17]	Compared to DP baselines [17]	Not specified [17]	Gradient/weight MIA [17]
Tianqu Kang et al., 2024	Yes	Theory + experiment [18]	Image (CIFAR-10) [18]	Quantized Gaussian mechanism [18]	Client-level [18]	LiRA MIA [18]
Michael Meyo et al., 2025	Yes	Comparative simulation [19]	Cross-border telemedicine (synthetic EHR) [19]	Client-side DP, $\epsilon = 0.30$ [19]	Record-level [19]	Model-inversion + MIA [19]
Zheng Zhang et al., 2023	No	Primary experiment [20]	Remote sensing image classification [20]	LDP with layer-wise noise [20]	Record-level (LDP) [20]	White-box MIA [20]
Yuke Hu et al., 2023	Yes	Attack + defense evaluation [21]	Knowledge graph embedding [21]	DP-Flames with private selection, $\epsilon = 16, 32$ [21]	Entity/triple-level [21]	Server-based and client-based triple MIA [21]
Abuzar Khan et al., 2026	Yes	Simulation study [3]	Maritime AIS vessel classification [3]	DP-FedAvg, $\sigma = 0.75-1.0$ [3]	Client-level [3]	MIA on global model [3]
Viswa Chaitanya Marella et al., 2025	Yes	Benchmark evaluation [22]	Oncologic CT tumor segmentation [22]	DP-SGD, $\sigma = 1.2, C = 1.0$ [22]	Client-level (approximate) [22]	MIA classifier [22]
Rui Liu et al., 2026	Yes	Decentralized FL experiment [23]	Image (Fashion-MNIST) [23]	Randomized-response LDP, $\epsilon = 0.5-8$ [23]	Record-level (LDP) [23]	MIA on local models [23]
Xi Zhou & Chunmei Yuan, 2026	Yes	Controlled FL simulation [24]	English learner corpus NLP [24]	Adaptive DP-SGD, $\epsilon = 1-8$ [24]	Record-level [24]	Shadow-model MIA [24]
Mohammad Naseri et al., 2020	Yes	Comprehensive DP evaluation [25]	Image, purchase, hospital records [25]	LDP ($\epsilon = 8.6$) and CDP ($\epsilon = 5.8$) [25]	Record-level (LDP), participant-level (CDP) [25]	White-box passive/active, global/local MIA [25]

Study	Full text retrieved?	Study type	Application domain	DP variant(s)	Privacy unit	Adversary model
Tonoy Kanti Chowdhury & Sai Praveen Kudapa, 2024	Yes	Multi-site comparative study [11]	Healthcare (imaging, EHR, wearable) [11]	DP + secure aggregation, $\epsilon = 3.0$ [11]	Record-level [11]	MIA re-identification [11]
Fan Xia et al., 2024	Yes	Attack method evaluation [26]	Image (CIFAR-10/100, MNIST) [26]	DP-SGD ($\epsilon \approx 40$) [26]	Record-level [26]	Label-only adversarial-perturbation MIA [26]
T. Jourdan et al., 2021	Yes	Comparative evaluation [27]	Motion sensor activity recognition [27]	LDP (Gaussian $N(0,0.01)$) [27]	Record-level (LDP) [27]	Server-side attribute + MIA [27]
Chetan Pathade & Shubham Patil, 2025	Yes	Attack-only evaluation [28]	Synthetic genomic data [28]	None applied [28]	N/A [28]	MIA, gradient-MIA, label inference [28]
M. O. Nassar & F. F. AL-Mashagba, 2025	Yes	Benchmark FL simulation [13]	Multimodal medical (ECG + clinical text) [13]	DP ($\epsilon = 1,5,10$), SA, hybrid DP+SA [13]	Record-level [13]	White-box MIA, gradient leakage, model inversion [13]
Shuangqing Xu et al., 2024	No	Framework + experiment [29]	Not specified (FL benchmark) [29]	Shuffle-model DP with RDP [29]	Record-level (shuffle) [29]	Not specified [29]
Jiaxin Li et al., 2024	Yes	Subject-level attack study [10]	Synthetic, Shakespeare, FEMNIST [10]	Item-level and subject-level DP [10]	Item-level and subject-level [10]	Server-side subject-level source inference [10]
Lin Chen et al., 2026	Yes	IDS framework evaluation [14]	Power network intrusion detection [14]	HE + calibrated Gaussian DP [14]	Client-level [14]	MIA + gradient leakage [14]
Jaba Tqemaladze, 2026	Yes	Platform validation [2]	Clinical AI (MIMIC-IV, eICU-CRD) [2]	DP-SGD, $\epsilon = 2.0/\text{round}$ [2]	Record-level [2]	MIA classifier [2]
Wei Yuan et al., 2023	Yes	Attack + defense study [12]	Federated recommendation [12]	LDP (Gaussian, $\lambda = 0.001-0.1$) [12]	Interaction-level [12]	Interaction-level MIA [12]
Judith Sáinz-Pardo Díaz et al., 2025	Yes	Comparative FL experiment [30]	Medical imaging [30]	Global DP and metric-privacy [30]	Client-level (server-side) [30]	Novel client-inference attack [30]

Study	Full text retrieved?	Study type	Application domain	DP variant(s)	Privacy unit	Adversary model
Mirwais Ahmadzai & Giang Nguyen, 2024	Yes	Empirical evaluation [31]	Public administration (payroll, opinion survey) [31]	Gaussian DP, noise multipliers 2–10 [31]	Client-level [31]	Server-side MIA [31]
Shaykat Purokayisto et al., 2025	Yes	FL simulation experiment [32]	Image (MNIST) [32]	Adaptive DP-SGD, $\epsilon = 0.66$ – 1.78 [32]	Record-level [32]	Not directly measured [32]
Mengyao Ma et al., 2023	No	Defense evaluation [33]	Benchmark datasets [33]	Server-side AGRs evaluated [33]	Record-level [33]	Poisoning MIA [33]
Zhenpeng Liu et al., 2022	No	Defense framework [34]	Not specified [34]	Gradient clipping + noise + trust domains [34]	Record-level [34]	White-box MIA [34]
Aron N. Horvath et al., 2023	Yes	Parameter-sweep evaluation [35]	Clinical ICU mortality (MIMIC-III) [35]	DP-SGD and DP-SVT [35]	Record-level [35]	Estimated privacy leakage (ϵ) [35]
Minh N. Vu et al., 2024	Yes	Active attack study on LLMs [7]	NLP (IMDB, Yelp, Twitter, Finance) [7]	GRR, RAPPOR, HE, dBit-FlipPM, $\epsilon = 5$ – 10 [7]	Record-level (LDP) [7]	Active dishonest server [7]
Hannes Hilberger et al., 2022	Yes	Comparative FL evaluation [36]	Histopathology (PathMNIST) [36]	Client-side DP, $\epsilon \approx 0.5$ [36]	Record-level [36]	Honest-but-curious MIA [36]
Ning Wang et al., 2022	Yes	Meta-learning DP framework [37]	Image (Omniglot, CIFAR-FS, Mini-ImageNet) [37]	DP-AGR/DP-AGRLR, $\epsilon = 1.5$ [37]	User-level and record-level [37]	Honest-but-curious server [37]
Rui Hu et al., 2021	Yes	Theory + experiment [38]	Tabular (Adult dataset) [38]	zCDP + secure aggregation [38]	Client-level [38]	Not directly measured (MIA) [38]
Soumya Banerjee et al., 2023	Yes	Attack evaluation [39]	Image (CIFAR-10/100, MNIST, FashionMNIST) [39]	FL as defense (no explicit DP) [39]	Record-level [39]	Shadow-model MIA [39]
Liwei Zhang et al., 2023	No	Attack method [40]	Four public datasets [40]	Tested against common defenses [40]	Record-level [40]	Local participant + global aggregator [40]
Tao Zhou, 2022	No	DP framework [41]	Not specified [41]	Hierarchical f-DP, Gaussian noise [41]	Client-level [41]	Not directly measured [41]

Study	Full text retrieved?	Study type	Application domain	DP variant(s)	Privacy unit	Adversary model
Jinnat Ara & Samiha Binte Abdullah, 2026	Yes	Retrospective multi-site study [42]	US healthcare diagnostics [42]	DP training, FL, hybrid [42]	Record-level [42]	MIA leakage evaluation [42]
Murali Krishna Pasupuleti, 2025	Yes	Benchmark experiment [43]	Precision medicine (WDBC) [43]	DP-style clipping + noise [43]	Record-level [43]	Loss-based MIA proxy [43]
Hybrid Differentially Private, 2026	Yes	Synthetic EHR generation [5]	Healthcare EHR (MIMIC-III, eICU) [5]	Local randomizer + DP-SGD, composed $\epsilon \leq 1.30$ [5]	Record-level [5]	Shadow, logistic, attribute MIA [5]
Wenbiao Li et al., 2026	No	Preprocessing verification [44]	Genomic data [44]	LDP, $\epsilon = 3$ [44]	Record-level [44]	MIA power evaluation [44]
Ras Dwivedi et al., 2026	Yes	DP ablation study [45]	Synthetic multimodal [45]	DP-SGD, $\epsilon = 1-8$ [45]	Record-level [45]	Black-box MIA [45]
Gongxi Zhu et al., 2024	Yes	MIA benchmark evaluation [46]	Image (CIFAR-100, DermNet, Tiny-ImageNet) [46]	Gradient perturbation, sparsification, etc. [46]	Record-level [46]	Multi-temporal, multi-spatial MIA [46]
A. Boutet et al., 2021	Yes	Defense evaluation [47]	Image and activity recognition [47]	Compared to LDP; MixNN proposed [47]	Record-level [47]	Server-side MIA + attribute inference [47]
Kongyang Chen et al., 2024	Yes	Attack evaluation [48]	Wearable HAR [48]	L2 regularization + dropout (no DP) [48]	Record-level [48]	Malicious server MIA [48]
Cédric Boscher et al., 2024	No	Defense middleware evaluation [49]	Six benchmark datasets [49]	Personalized FL (DINAR) [49]	Record-level [49]	MIA against FL models [49]
I. Shiri et al., 2024	Yes	Multi-site clinical study [1]	COVID-19 CT prognostication [1]	GDP-AQuCl (DP + adaptive clipping) [1]	Record-level [1]	MIA evaluated per epoch [1]
Baoping Wang et al., 2024	Yes	Federated object detection [50]	Object detection (COCO, PASCAL VOC) [50]	Dynamic DP, $\epsilon = 0.5-1.0$ [50]	Record-level [50]	CNN/RNN MIA models [50]
K. Bharath et al., 2026	Yes	Hybrid privacy framework [51]	Healthcare (MIMIC-III, COVID X-ray) [51]	DP + HE hybrid, $\epsilon = 0.7$ [51]	Record-level [51]	MIA + model inversion [51]

Study	Full text retrieved?	Study type	Application domain	DP variant(s)	Privacy unit	Adversary model
Xiyuan Yang et al., 2025	Yes	Federated clustering [52]	Image and text (EMNIST, CIFAR) [52]	Client-level DP with RR-Cluster [52]	Client-level [52]	Privacy-utility analysis [52]
Mónica Ribero et al., 2020	Yes	Recommendation framework [53]	Medical (eICU), movie ratings [53]	ϵ -DP prototypes, $\epsilon = 0.1$ [53]	Record-level [53]	Formal DP guarantee only [53]
Alexander Shen et al., 2022	Yes	Privacy-utility study [54]	Mobile health (mood prediction) [54]	DP with gradient clipping [54]	Record-level [54]	Simulated property inference [54]
Mahavir Teraiya & Madhu Shukla, 2026	Yes	Systems-design evaluation [55]	IIoT intrusion detection [55]	Client-level DP, $\epsilon = 0.5-10$ [55]	Client-level [55]	Privacy-utility frontier (no direct MIA) [55]
Kongyang Chen et al., 2025	No	Attack method [56]	FCL across diverse datasets [56]	None (attack-focused) [56]	Record-level [56]	Client-side passive/active MIA [56]
Haley Hoech et al., 2022	Yes	One-shot FL evaluation [57]	Image (CIFAR-10) [57]	Full (ϵ, δ) -DP, $\epsilon = 0.1-0.5$ [57]	Record-level [57]	No direct MIA [57]
G. Saxena, 2026	No	Confidential-computing framework [58]	Image, NLP, clinical (MIMIC-III) [58]	DP + TEE [58]	Record-level [58]	MIA over 10,000 queries [58]
Yiwei Li et al., 2025	Yes	Model-splitting framework [59]	Tabular (Adult), image (MNIST) [59]	DP on public submodel only [59]	Record-level (LDP) [59]	Privacy amplification analysis [59]
Amey Mirashi et al., 2026	Yes	FL framework evaluation [60]	Cardiovascular (Framingham) [60]	SecAgg+ plus Central DP [60]	Record-level [60]	No empirical MIA reported [60]
Xue Yang et al., 2024	Yes	Bidirectional privacy scheme [61]	Image (CIFAR-10/100), medical [61]	MP_Server + DDP_Client (LDP) [61]	Record-level [61]	MIA + reconstruction attack [61]
Yiwei Li et al., 2022	Yes	Theory + experiment [62]	Tabular (Adult) [62]	FedSPD-DP (ϵ, δ) -DP [62]	Client-level [62]	Privacy-utility tradeoff [62]
Milad Nasr et al., 2018	Yes	Comprehensive MIA study [63]	Image (CIFAR-100), purchase, hospital [63]	Not applied as defense [63]	Record-level [63]	White-box passive/active, global/local [63]
Zheng Xu et al., 2023	Yes	Production deployment [64]	Gboard language models (NWP, OTF) [64]	DP-FTRL, ρ -zCDP, $\rho \in (0.3, 2)$ [64]	Device-level [64]	No direct MIA reported [64]

Study	Full text retrieved?	Study type	Application domain	DP variant(s)	Privacy unit	Adversary model
Ozer Ozturk et al., 2025	Yes	FL simulation [65]	Hypertension classification [65]	Randomized response, $\epsilon = 0-5$ [65]	Record-level [65]	Privacy-performance tradeoff [65]
Hui-Po Wang et al., 2023	Yes	FL optimization method [66]	Image (MNIST, FashionMNIST, CIFAR-10) [66]	Record-level DP, $\epsilon = 2.79, 10.18$ [66]	Record-level [66]	No empirical MIA [66]
Devharsh Trivedi et al., 2026	Yes	Split-learning framework [67]	Log classification (HDFS), image (MNIST) [67]	Noise flooding (no formal DP) [67]	Not formal DP [67]	Shadow-model MIA [67]
Dmitrii Usynin et al., 2021	Yes	Model-adaptation study [68]	Image (CIFAR-10, MNIST), medical imaging [68]	DP-SGD, $\epsilon = 6.0$ [68]	Record-level [68]	MIA + reconstruction attacks [68]
Omer Haliloglu et al., 2021	No	FL + DP evaluation [69]	Mobile network RSRP estimation [69]	DP (details not reported) [69]	Not specified [69]	MIA threat model [69]
Natalie Lang et al., 2022	Yes	Joint privacy + quantization [70]	Image (MNIST, CIFAR-10) [70]	JoPEQ ϵ -LDP [70]	Record-level (LDP) [70]	Gradient-inversion attack [70]
Feng Guo et al., 2025	No	Adaptive DP framework [71]	Image (MNIST, FMNIST, CIFAR-10) [71]	Adaptive personalized DP [71]	Record-level [71]	Not specified [71]
Zhenxiao Zhang et al., 2024	Yes	DPFL method evaluation [72]	Image (FEMNIST, CIFAR-10) [72]	DP ² -FedSAM, $\epsilon = 1.0$ [72]	Client-level [72]	No direct MIA measured [72]
Shuya Feng et al., 2024	Yes	DP-FL framework + attacks [73]	Image (MNIST, Medical MNIST, CIFAR-10, UTKFace) [73]	UDP-FL (Staircase, Gaussian, Laplace), $\epsilon = 2-8$ [73]	Record-level [73]	Shokri, LiRA, Canary MIA [73]
Xue Jiang et al., 2022	No	LDP-FL framework [74]	Structured + image datasets [74]	LDP with dimension selection [74]	Record-level (LDP) [74]	Not specified [74]
Zi Ye et al., 2026	Yes	Shuffle-model DP evaluation [75]	Image (MNIST, CIFAR-10/100) [75]	AdaCOS shuffle DP, $\epsilon_1 = 4000$ [75]	Record-level (shuffle) [75]	No empirical MIA [75]

Study	Full text retrieved?	Study type	Application domain	DP variant(s)	Privacy unit	Adversary model
Roman Schefzik et al., 2025	No	Federated MTL with DP [76]	Gene expression (schizophrenia) [76]	Laplace mechanism [76]	Record-level [76]	MIA threat described [76]
Xiaoyi Lin & Guangrong Duan, 2025	Yes	Blockchain + DP + FL [77]	Healthcare IoT (MIMIC-III, ChestX-ray14) [77]	Adaptive DP, $\epsilon = 0.5-5.0$ [77]	Record-level [77]	MIA success rate [77]
Hojat Allah Salehi et al., 2024	Yes	DP mechanism evaluation [78]	Image (MNIST, CIFAR-10) [78]	CorBin-FL PLDP [78]	Parameter-level LDP [78]	Privacy-utility analysis [78]
Faeze Moradi Kalarde et al., 2025	Yes	OTA FL with DP [79]	Image (MNIST, CIFAR-10) [79]	Adaptive RDP [79]	Device-level [79]	RDP/DP leakage metrics [79]
Luca Melis et al., 2018	Yes	Comprehensive inference attack study [80]	Image, location, text, face [80]	Record-level and participant-level DP discussed [80]	Record- and participant-level [80]	Passive/active collaborative-learning MIA + property inference [80]

The included studies span diverse federated-learning configurations: centralized-server FL (the dominant architecture), decentralized/peer-to-peer FL [23], split learning [67], federated recommender systems [12], federated knowledge graph embedding [21], and federated contrastive learning [56]. Application domains include healthcare and clinical AI (approximately 25 studies), computer vision benchmarks (approximately 25 studies), natural language processing (approximately 8 studies), IoT and network security (approximately 6 studies), and specialized domains such as maritime intelligence [3], public administration [31], remote sensing [20], and genomics [28, 44]. DP variants range from local differential privacy (LDP), central differential privacy (CDP), and DP-SGD to more specialized approaches including Rényi DP accounting [4], zero-concentrated DP [38], f-DP [41], metric-privacy [30], and shuffle-model DP [29, 75].

Quantitative Effects of Differential Privacy on Membership Inference

The following table reports empirical MIA results from studies that provide quantitative MIA metrics both with and without DP, or under multiple DP configurations. MIA success is reported using the metric specified in each study (attack accuracy, AUC, F1, or advantage), alongside the corresponding model utility and epsilon value.

Study	Domain	ϵ	MIA without DP	MIA with DP	Metric	Utility without DP	Utility with DP
Taslina Khanam et al., 2026	EEG-BCI	5	Limited intrinsic FL protection [15]	Substantially reduced, especially gradient- and identity-level [15]	Attack success	Baseline accuracy [15]	Best privacy-utility balance [15]

Study	Domain	ϵ	MIA without DP	MIA with DP	Metric	Utility without DP	Utility with DP
Truc Nguyen et al., 2023	CIFAR-10, ImageNet, CelebA	5 (LDP)	Near 100% success [6]	≥ 0.77 success at $\epsilon \geq 5$ [6]	Attack success rate	Baseline FL accuracy [6]	Severely damaged at small ϵ [6]
A. M & C. L. Chowdhary, 2026	Medical MRI	1.5	Not reported separately	52.1% attack accuracy [16]	Attack accuracy	FedAvg 86.5% [16]	97.3% accuracy [16]
Samuel Maddock et al., 2022	CIFAR-10, CelebA, text	Formal $\epsilon \approx 50$	N/A (canary audit)	Empirical $\hat{\epsilon} \approx 6$ (CelebA) [8]	Empirical ϵ	N/A	No accuracy impact [8]
Adiah Qazi et al., 2026	ERP fraud	(8,1.0) RDP	Not reported	51.2% shadow MIA, 53.8% LiRA [4]	Attack success %	N/A	93.5% F1 [4]
Anshuman Suri et al., 2022	FEMNIST	4.0	F1 = 0.83 [9]	Item DP: 0.76; User DP: 0.67; Subject DP: 0.72 [9]	Attack F1	91.9% accuracy [9]	85.1% (item), 41.0% (user), 81.5% (subject) [9]
Tianqu Kang et al., 2024	CIFAR-10	Varies	Higher MIA accuracy (unquantized) [18]	Lower MIA accuracy (quantized) [18]	MIA accuracy	Higher [18]	Lower with quantization [18]
Michael Meyo et al., 2025	Telemedicine EHR	0.30	MIA AUC ≈ 0.50 (all modes) [19]	MIA AUC ≈ 0.50 [19]	AUC	FL: 0.682 accuracy [19]	FL+DP: 0.530 accuracy [19]
Abuzar Khan et al., 2026	Maritime AIS	$\sigma \leq 0.5$	AUC ≈ 0.51 (FL alone) [3]	AUC ≈ 0.51 , zero correct inferences [3]	ROC AUC	Val AUC ≈ 0.90 [3]	$\sim 98\%$ utility retained [3]
Viswa Chaitanya Marella et al., 2025	Tumor segmentation	~ 10 (per round)	AUC 0.68–0.72 [22]	DP-SGD: AUC 0.25 [22]	MI AUC	Dice 0.85 [22]	Dice 0.79 [22]
Rui Liu et al., 2026	Fashion-MNIST	0.5–8	Baseline attack AUC varies [23]	AUC 0.497 (DeFL-VP) [23]	Attack AUC	Varies [23]	88.7% accuracy [23]
Xi Zhou & Chunmei Yuan, 2026	English learner NLP	2	Not reported	MIA AUC 0.521 [24]	ROC AUC	Centralized baseline [24]	GED F1 0.809, WPA 74.2% [24]
Mohammad Naseri et al., 2020	CIFAR-100	LDP 8.6; CDP 5.8	75% (white-box active) [25]	LDP: 55%; CDP: 52% [25]	Attack accuracy	Baseline [25]	Reduced [25]

Study	Domain	ϵ	MIA without DP	MIA with DP	Metric	Utility without DP	Utility with DP
Tonoy Kanti Chowdhury & Sai Praveen Kudapa, 2024	Healthcare	3.0	42.5% advantage (centralized) [11]	8.6% advantage (DP+SA) [11]	MIA advantage	AUROC 0.936 [11]	AUROC 0.917–0.931 [11]
Fan Xia et al., 2024	CIFAR-10	~40	63.0% accuracy [26]	DP-SGD: 56.0% [26]	Attack accuracy	Baseline [26]	Reduced [26]
T. Jourdan et al., 2021	Motion sensors	N/A (Gaussian noise)	FL: membership near random [27]	FedPer: strongest MIA protection [27]	MIA accuracy	FL baseline [27]	FedPer slightly improved [27]
M. O. Nassar & F. AL-Mashagba, 2025	Medical multimodal	1–10	Baseline MIA success [13]	Hybrid DP+SA: MIA 9.4% [13]	Attack success %	Baseline AUROC [13]	Early fusion AUROC 0.869 [13]
Jiaxin Li et al., 2024	Synthetic, Shakespear	Item DP, Subject DP	0.888 accuracy (Synthetic) [10]	Subject DP: 0.68 (Shakespear) [10]	Attack accuracy	Baseline [10]	Utility impaired [10]
Lin Chen et al., 2026	Power network IDS	$\epsilon_0 = 0.5$	FedAvg advantage 0.234 [14]	HE+DP: 0.078 [14]	MIA advantage	FedAvg F1 0.8828 [14]	Hybrid F1 0.9450 [14]
Jaba Tqemaladze, 2026	Clinical AI (MIMIC-IV)	2.0/round	AUC 0.61 $\pm$ 0.04 [2]	AUC 0.52 $\pm$ 0.03 [2]	MIA AUC	Oracle AUC 0.789 [2]	FCLC AUC 0.758 [2]
Wei Yuan et al., 2023	Recommender systems	LDP $\lambda = 0.001$ –0.1	IMIA outperforms random [12]	LDP: ineffective unless utility destroyed [12]	F1	Hit@10 baseline [12]	Reduced Hit@10 [12]
Mirwais Ahmadzai & Giang Nguyen, 2024	Public administration	0.1	Higher MIA accuracy/F1 [31]	Declining with noise multiplier [31]	MIA accuracy, F1	Baseline accuracy [31]	Stable utility [31]
Jinnat Ara & Samiha Binte Abdullah, 2026	US healthcare	Not specified	Baseline leakage [42]	DP: –0.083 leakage; hybrid: –0.071 [42]	Leakage reduction (β)	AUROC baseline [42]	Δ AUC –0.012 (DP), –0.004 (FL) [42]
Murali Krishna Pasupuleti, 2025	Precision medicine	Illustrative	Centralized AUC 0.5486 [43]	FedAvg+DP: AUC 0.5177 [43]	MIA AUC	Centralized AUROC 0.9964 [43]	DP AUROC 0.9924 [43]

Study	Domain	ϵ	MIA without DP	MIA with DP	Metric	Utility without DP	Utility with DP
Hybrid Differentially Private, 2026	Synthetic EHR	≤ 1.30	N/A (generative model)	Shadow 0.503, Logistic 0.512, Attribute 0.498 [5]	MIA AUROC	N/A	AUROC 0.884 [5]
Wenbiao Li et al., 2026	Genomics	3	N/A	MIA power < 0.05 [44]	MIA power	N/A	80% verification accuracy [44]
Ras Dwivedi et al., 2026	Synthetic multimodal	1–8	Not applicable (DP always on) [45]	0.507 at $\epsilon=1$; 0.482 at $\epsilon=8$ [45]	MIA accuracy	0.689 at $\epsilon=1$ [45]	0.754 at $\epsilon=8$ [45]
I. Shiri et al., 2024	COVID-19 CT	Controlled by DP	N/A	AUC 0.50 ± 0.003 (range 0.49–0.51) [1]	MIA AUC	Centralized: acc 0.76 [1]	DPFL: acc 0.75 [1]
Baoping Wang et al., 2024	Object detection	0.5, 1.0	Higher success (non-IID, RNN) [50]	Lower at $\epsilon = 0.5$ [50]	Attack success %	Full mAP [50]	Reduced mAP [50]
K. Bharath et al., 2026	Healthcare	0.7	FL only: 68% resistance [51]	Hybrid: 92% resistance [51]	MIA resistance	FL only: 89.0% [51]	Hybrid: 93.0% [51]
Alexander Shen et al., 2022	Mobile health	Varied σ	$> 90\%$ leakage risk (FL only) [54]	59.6% attack success [54]	Attack success %	Baseline R^2 [54]	R^2 reduced by 10pp [54]
G. Saxena, 2026	Image, NLP, clinical	Not specified	71.3% success (un-protected) [58]	51.8% success [58]	Attack success %	Baseline [58]	11.9–13.9% latency overhead [58]
Xue Yang et al., 2024	CIFAR-10/100	Small	Not reported	$\sim 50\%$ ASR (random guessing) [61]	ASR %	Privacy-free baseline [61]	$< 6\%$ accuracy loss [61]
Xiaoyi Lin & Guangrong Duan, 2025	Healthcare IoT	1.0	Vanilla FL baseline [77]	15.6% MIA success (78.5% reduction) [77]	Attack success %	Vanilla FL accuracy [77]	94.3% accuracy [77]
Shuya Feng et al., 2024	CIFAR-10	2–8	NbAFL and DP-SGD baselines [73]	Staircase UDP-FL: lowest MIA leakage [73]	TPR at FPR=0.01, AUC	Baseline accuracy [73]	$\sim 90\%$ MNIST in 25 epochs [73]

Study	Domain	ϵ	MIA without DP	MIA with DP	Metric	Utility without DP	Utility with DP
Minh N. Vu et al., 2024	NLP (LLMs)	5–10	100% success (FC attack, no DP) [7]	AUC < 0.72 at $\epsilon = 5$ [7]	Attack AUC	Baseline LLM [7]	DP-protected LLM [7]
Dmitrii Usynin et al., 2021	CIFAR-10	6.0	59.8% MIA accuracy [68]	54.3% with DP [68]	MIA accuracy	77.2% model accuracy [68]	71.5% model accuracy [68]
Kongyang Chen et al., 2024	Wearable HAR	None applied	Up to 92% MIA accuracy [48]	L2+dropout: minimally effective [48]	MIA accuracy	Baseline HAR [48]	Minimal change [48]
Luca Melis et al., 2018	Location, image, text	Record/participant DP discussed	Precision 0.92–0.99 [80]	10% params: AUC 0.84–0.94 [80]	Precision, AUC	Baseline [80]	Participant DP: no convergence with ≤ 30 participants [80]

Several patterns emerge from the quantitative evidence. When DP reduces MIA attack AUC or accuracy to 0.50–0.53, the empirical leakage is considered indistinguishable from random guessing, which is the strongest empirical privacy outcome achievable. This near-chance MIA performance was reported across diverse domains: COVID-19 CT prognostication (AUC 0.50 ± 0.003) [1], maritime vessel classification (AUC ≈ 0.51) [3], ERP fraud detection (51.2%) [4], clinical AI with DP-SGD (AUC 0.52 ± 0.03) [2], medical MRI classification (52.1%) [16], synthetic EHR generation (AUROC 0.503) [5], and power network intrusion detection (advantage 0.078) [14]. In most of these cases, the epsilon values were moderate ($\epsilon = 0.5$ – 2.0) and the model utility loss was contained within 1–5 percentage points of AUC or accuracy relative to non-private baselines.

However, a distinct group of studies reported that DP provided only partial MIA mitigation, with attack success remaining substantially above chance. Active server-side attacks against LDP-protected FL achieved success rates of at least 0.77 at $\epsilon \geq 5$ [6], and even at very small ϵ the attack remained above random guessing while severely degrading model utility [6]. Similarly, the adversarial-perturbation-based MAPD_MIA attack on DP-SGD-protected models retained accuracy of 56.0% on CIFAR-10 and 58.0% on MNIST, above the 50% baseline [26]. In federated recommendation, LDP was not effective against interaction-level MIA unless recommendation performance was substantially compromised [12]. Subject-level attacks on FEMNIST still achieved F1 of 0.67–0.76 under various DP granularities at $\epsilon = 4.0$ [9].

Variation by Privacy Unit

A critical dimension of heterogeneity across studies is the granularity at which DP protections operate. Three principal privacy units are examined: record-level (item-level), client-level (user/participant-level), and subject-level.

Anshuman Suri et al. (2022) provided the most systematic comparison across privacy units within a single study on FEMNIST. Under $\epsilon = 4.0$, item-level DP reduced subject MIA F1 from 0.83 to 0.76 while maintaining 85.1% task accuracy; user-level DP reduced MIA F1 more aggressively to 0.67 but at the cost of task accuracy dropping to 41.0%; and subject-level DP achieved intermediate MIA F1 of 0.72 with 81.5% task accuracy [9]. This demonstrates that user-level DP provides the strongest MIA mitigation but with the most severe utility degradation, while subject-level DP offers a more favorable privacy-utility trade-off for the cross-silo setting where data subjects may span multiple participating organizations.

Jiaxin Li et al. (2024) similarly compared item-level and subject-level DP against the SLSIA attack, finding that subject-level DP generally reduced SLSIA more effectively than item-level DP. Nevertheless, the attack remained

effective in several configurations — for example, accuracy of 0.68 on Shakespeare under subject-level DP and 0.58 under item-level DP — and both DP variants could substantially impair model utility [10].

At the client-level granularity, several studies demonstrated strong privacy outcomes. The FCLC platform operating at $\epsilon = 2.0$ per round achieved MIA AUC of 0.52 ± 0.03 , indistinguishable from chance [2]. Mahavir Teraiya and Madhu Shukla (2026) swept ϵ from 0.5 to 10 at the client level, observing macro-F1 declining from 0.93 to 0.78 as the budget tightened [55]. MM-PFL-ADP explicitly noted that its formal privacy guarantee protects entire clients rather than individual records, yet membership-inference accuracy still fell to 52.1% at $\epsilon = 1.5$ [16].

The distinction between record-level and client-level DP has practical implications highlighted by Melis et al. (2018), who noted that record-level DP bounds membership inference for individual records but does not prevent group property inference, whereas participant-level DP theoretically bounds all attack types but failed to produce a converged model with up to 30 participants in their experiments [80]. Naseri et al. (2020) confirmed this empirically: LDP (record-level) and CDP (participant-level) both reduced white-box MIA accuracy on CIFAR-100 from 75% to 55% and 52% respectively, but neither defended against property inference [25].

The canary-audit approach of CANIFE measured privacy at the record level and found that empirical per-round epsilon was 4–5 times lower than the theoretical bound [8], suggesting that formal DP guarantees may be substantially conservative for record-level privacy under realistic FL threat models.

Variation by Adversary Access to Model Updates

The degree of adversary access to model updates is a major determinant of MIA effectiveness and, correspondingly, of the level of DP protection required.

Passive black-box adversaries, who observe only the final global model or its predictions, represent the weakest threat model. Under this setting, several studies reported that even standard FL without DP provided substantial privacy: MIA AUC remained near 0.50 for cross-border telemedicine [19], maritime classification [3], and precision medicine [43]. DP further reduced residual leakage modestly in these cases.

Passive white-box adversaries, who access model parameters or gradients, achieve substantially higher MIA success. Nasr et al. (2018) demonstrated that white-box access increased membership-inference accuracy from 54.5% (black-box) to 74.3% on DenseNet [63]. In federated settings, a passive global attacker reached 79.2% on CIFAR-100 DenseNet [63]. Naseri et al. (2020) showed white-box MIA accuracy of 75% on CIFAR-100 without defenses [25], which LDP and CDP reduced to 55% and 52% respectively [25].

Active adversaries — who manipulate model parameters or training — represent the most challenging threat. Truc Nguyen et al. (2023) showed that a dishonest server performing active membership inference achieved near-100% success without LDP and maintained success rates ≥ 0.77 even under LDP at $\epsilon \geq 5$ [6]. Minh N. Vu et al. (2024) found that active attacks on LLMs achieved 100% success when targeting fully connected layers without DP; at $\epsilon = 5$, nearly all attacks fell below AUC 0.72, but at weaker privacy regimes significant leakage persisted [7]. Nasr et al. (2018) showed that combining participant isolation with gradient ascent pushed federated MIA accuracy to 87.3% on CIFAR-100 [63].

Multi-temporal and multi-spatial adversaries compound the risk. Gongxi Zhu et al. (2024) found that combining model information across multiple communication rounds (multi-temporal) and from non-target clients (multi-spatial) substantially improved MIA effectiveness, especially under homogeneous client data [46]. This observation implies that defenses must account for cumulative information leakage across rounds, not just single-round exposure.

Server-side versus client-side adversaries also matter. Kongyang Chen et al. (2024) demonstrated that a curious malicious server achieved up to 92% MIA accuracy on wearable HAR datasets [48], and L2 regularization plus dropout provided minimal protection [48]. Conversely, client-side attacks in federated contrastive learning were effective even without server manipulation capability [56]. The LoDen defense specifically targeted poisoning MIA from malicious clients, achieving high detection rates [33], while server-side robust aggregation algorithms were largely insufficient against such targeted attacks [33].

Variation by Application Domain

Healthcare and Clinical AI

Healthcare applications constitute the largest domain cluster and generally report favorable DP outcomes. The DPFL model for COVID-19 CT prognostication across 19 centers maintained accuracy of 0.75 versus 0.76 centralized (DeLong $p = 0.98$), with MIA AUC of 0.50 ± 0.003 [1]. The FCLC platform on MIMIC-IV clinical data achieved AUC 0.758 with MIA AUC 0.52 under DP [2]. Tonoy Kanti Chowdhury and Sai Praveen Kudapa (2024) reported MIA advantage declining from 42.5% centralized to 8.6% with DP+SA across imaging, EHR, and wearable datasets [11]. Jinnat Ara and Samiha Binte Abdullah (2026) found that DP training reduced membership-inference leakage by $\beta = -0.083$ with only $\Delta\text{AUC} = -0.012$ [42]. The B-DPFL framework achieved 94.3% accuracy with MIA success reduced to 15.6% at $\epsilon = 1.0$ [77]. The hybrid DP+HE framework on MIMIC-III achieved 93.0% accuracy with 92% MIA resistance [51]. In tumor segmentation, DP-SGD drove MIA AUC from 0.68–0.72 down to 0.25, but Dice score fell from 0.85 to 0.79 [22]. For multimodal medical data, hybrid DP+SA reduced MIA success to 9.4% with AUROC improvements from cross-modal fusion [13]. These results suggest that healthcare applications, where datasets tend to be moderately sized and models are relatively simple classifiers or segmentation networks, allow DP to achieve near-chance MIA with modest utility costs.

Computer Vision Benchmarks

Results on standard vision benchmarks are more heterogeneous. On CIFAR-100, white-box MIA accuracy reached 75% without DP and was reduced to 52–55% by CDP/LDP [25]. On CIFAR-10, the adversarial-perturbation attack MAPD_MIA retained 56.0% accuracy even under DP-SGD [26], while DeFL-VP achieved attack AUC of 0.497 with 88.7% model accuracy [23]. DP²-FedSAM achieved 84.1% accuracy on CIFAR-10 at $\epsilon = 1.0$ but did not directly measure MIA [72]. FedLAP-DP achieved 71.91% CIFAR-10 accuracy under DP without reporting empirical MIA [66]. The FedOnco-Bench showed that DP-SGD reduced MIA AUC from 0.72 to 0.25 on synthetic CT data at the cost of 6 Dice points [22]. The CANIFE canary audit on CIFAR-10 estimated empirical per-round ϵ approximately 4–5 times lower than the theoretical bound [8].

Natural Language Processing

NLP applications showed both extreme vulnerability and effective protection depending on adversary capability. Without DP, active attacks on BERT, RoBERTa, DistilBERT, and GPT models achieved 100% success when targeting fully connected layers [7]. At $\epsilon = 5$, nearly all attacks fell below AUC 0.72 [7]. Google's production Gboard deployment demonstrated that DP-FTRL with ρ -zCDP guarantees ($\rho \in (0.3, 2)$) maintained high utility in live tests while providing formal privacy [64]. AdaDP-FedSec for English learner corpus training reduced MIA AUC to 0.521 at $\epsilon = 2$ [24].

IoT, Network Security, and Specialized Domains

In IoT intrusion detection, client-level DP with robust aggregation showed macro-F1 ranging from 0.78 to 0.93 across $\epsilon = 0.5$ –10 [55]. The power-network IDS framework achieved MIA advantage of 0.078 (versus 0.234 for FedAvg) while maintaining F1 = 0.9450 [14]. In maritime AIS classification, MIA was reduced to near-chance (AUC ≈ 0.51) with moderate DP noise [3]. For public administration payroll and opinion data, increasing Gaussian noise systematically reduced MIA success [31]. In federated recommendation systems, LDP was notably ineffective against interaction-level MIA unless recommendation quality was largely sacrificed [12]. In genomic applications, MIA power was kept below 0.05 at $\epsilon = 3$ [44], and gradient-based MIA on federated genomic data achieved the highest attack effectiveness (F1 = 0.87) without any DP protection [28].

Additional Defense Mechanisms and Comparisons

Several studies compared DP against alternative or complementary defenses. MixNN achieved MIA reduction of up to 73.9% versus classical FL without utility loss, outperforming both model compression and LDP [47]. The Digestive Neural Network achieved 16.17% higher classification accuracy and 9% lower attack accuracy than DP schemes under gradient-sharing FL [17]. DINAR reduced MIA success to its optimal value without accuracy loss or computational overhead, in contrast to existing defenses that incurred up to 35% client-side and 3,000% server-side overhead [49].

Personalized layers (FedPer) provided the strongest membership-inference protection on motion-sensor data, reaching near-random-guess accuracy, and outperformed LDP while also improving convergence [27]. Data Replacement mechanisms achieved the most favorable privacy-utility tradeoff compared to gradient perturbation approaches in the comprehensive evaluation by Gongxi Zhu et al. (2024) [46]. SplitML's model partitioning reduced MI member-accuracy to approximately 37% for top layers and 19% for bottom layers without formal DP accounting [67].

Combining DP with other mechanisms generally yielded stronger protection. Hybrid DP+SA reduced attack success by over 84% relative to baseline in multimodal medical FL [13]. The HE+DP combination in power-network IDS reduced MIA advantage to 0.078 versus 0.187 for DP-only [14]. Blockchain-enhanced DP reduced MIA success to 15.6% while achieving 94.3% accuracy [77]. CC-Shield's five-layer architecture combining TEE, DP, attestation, and encryption reduced MIA to 51.8% [58].

Synthesis

The apparent contradictions in MIA outcomes across studies — ranging from near-perfect defense (AUC 0.50) to persistent vulnerability (attack success ≥ 0.77) — can be largely reconciled by systematically examining three interacting factors: adversary capability, privacy budget magnitude, and application-specific data characteristics.

Adversary capability as the primary moderator. Studies reporting DP as highly effective against MIA (achieving near-chance attack performance) predominantly evaluated passive or honest-but-curious adversaries with black-box access to the final global model [1–5]. In contrast, studies reporting residual leakage despite DP employed active adversaries who could manipulate model parameters [6, 7] or white-box adversaries with access to per-round gradient updates [25, 63]. The Truc Nguyen et al. (2023) result — where active server-side attacks maintained ≥ 0.77 success at $\epsilon \geq 5$ under LDP — is not contradicted by the Shiri et al. (2024) finding of AUC 0.50 under DPFL, because the latter evaluated only post-training black-box MIA rather than active manipulation of model weights during training. Both findings are simultaneously valid within their respective threat models. The implication is that DP's protective value depends critically on the adversary assumed: for passive post-training evaluation, moderate DP suffices; for active in-training adversaries, even aggressive DP may be insufficient.

Privacy budget thresholds. Across the literature, a consistent pattern emerges in which MIA effectiveness drops to near-chance levels below certain epsilon thresholds, which vary by domain and model complexity. For healthcare classification models, $\epsilon \leq 2.0$ generally produced MIA AUC of 0.50–0.53 [1, 2, 19]. For image classification on CIFAR-scale tasks, $\epsilon = 5$ was generally sufficient under DP-SGD to bring active LLM attacks below AUC 0.72 [7], while stricter budgets ($\epsilon \leq 1.0$) were needed for stronger passive guarantees on tumor segmentation [22]. The non-linear relationship between ϵ and MIA success is evident: the DP ablation study by Ras Dwivedi et al. (2026) found MIA accuracy essentially flat across $\epsilon = 1$ –8 (ranging only from 0.507 to 0.482), suggesting that beyond a certain noise threshold the marginal privacy gain from tighter budgets is minimal [45]. This contrasts with utility, which showed clear degradation at small ϵ (0.689 accuracy at $\epsilon = 1$ versus 0.754 at $\epsilon = 8$) [45], indicating that practical deployment should target the inflection point where MIA reaches near-chance levels, rather than pursuing the tightest possible budget.

Domain-specific data characteristics. Healthcare applications consistently showed more favorable privacy-utility tradeoffs, likely because clinical prediction tasks use moderately complex models on structured or lower-dimensional data where DP noise has less relative impact on utility. In contrast, high-dimensional image or language tasks require larger models whose gradient updates carry more information, making them both more vulnerable to sophisticated MIA and more sensitive to DP noise. Federated recommendation systems represent a particularly challenging domain where interaction-level MIA exposed vulnerabilities that LDP could not mitigate without destroying recommendation quality [12]. Suri et al. (2022) found that attack effectiveness varied dramatically by domain even within a single experimental framework — near-random on Shakespeare but up to $F1 \approx 0.95$ on high-dimensional synthetic configurations — with higher vulnerability associated with greater data dimensionality, Dirichlet sampling, larger models, and longer training [9].

Privacy-unit mismatch. A frequently overlooked source of heterogeneity is the gap between the DP privacy unit and the actual inference target. Record-level DP protects individual training records but leaves subject-level patterns — which may span multiple records across multiple organizations — partially exposed [9, 10]. Client-level DP protects entire participant contributions but is more disruptive to model utility [9, 80]. Subject-level DP, while conceptually the most aligned with real-world privacy needs in cross-silo FL, remains under-studied relative to item-level and

client-level alternatives. The finding that user-level DP reduced FEMNIST accuracy to 41.0% (versus 85.1% for item-level DP at the same ϵ) [9] underscores that stronger privacy granularity comes at substantial utility cost, and practical deployments must carefully match the privacy unit to the actual threat.

Empirical versus theoretical privacy guarantees. The CANIFE canary-audit methodology revealed that empirical privacy loss was 4–5 times lower than the theoretical DP bound [8], and other studies similarly found that formal ϵ values substantially overestimate actual information leakage [8]. This suggests that the conservative threat model underlying DP theory — which assumes worst-case datasets and adversaries — may yield vacuous guarantees in practice, and that empirical auditing with canaries or shadow-model attacks provides more actionable privacy measurements for deployed systems.

In summary, DP reliably defends against passive membership inference across most domains when the privacy budget is appropriately calibrated (typically $\epsilon \leq 2$ –5 for healthcare, $\epsilon \leq 5$ –10 for vision and NLP). Against active adversaries who manipulate training, DP alone is insufficient and must be combined with secure aggregation, robust aggregation, or architectural defenses such as model splitting or layer mixing. The privacy unit should be matched to the inference target: record-level DP is adequate for record-level MIA, but cross-silo settings where subject data spans multiple organizations require subject-level or client-level protections that incur greater utility costs.

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